

DASC7606 – 6A

Deep Learning

Stable Diffusion Adversarial Images Reflections

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Topics

Stable Diffusion

Adversarial Images

- Neural Network
- Other Models

Reflections

- Distributed and Deep representations
- Generalization and Multi-task learning
- Myth of local minima

What is Stable diffusion ?

R. Rombach, et.al., High-Resolution Image Synthesis with Latent Diffusion Models, 2021

Stable diffusion - a scientific model about how things spread/move in a **stable** manner (same speed, steady)

- How a drop of ink in a glass of water spread from a concentrated spot to evenly throughout water
- People exiting a stadium after a game
- Heat, smell, information (gossip) spread

Diffusion Model - physics-inspired model



- forward diffusion
gradually adds noise with controlled amount $\beta_1 < \beta_2 < \dots < \beta_T < 1$
to clear image to turn it into noisy image

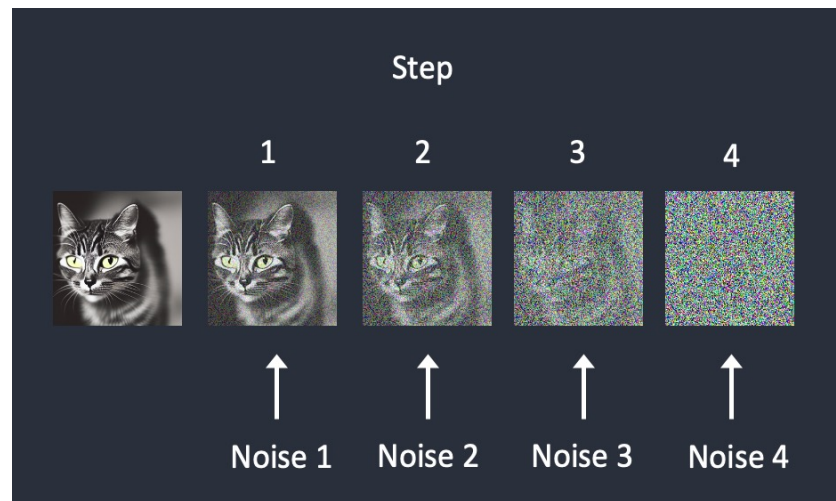
For synthesizing an image.

- reverse diffusion is to remove noise for generating the images

Diffusion model – Training Noise Predictor

Training - teach a NN to predict the noise added (**noise predictor**)

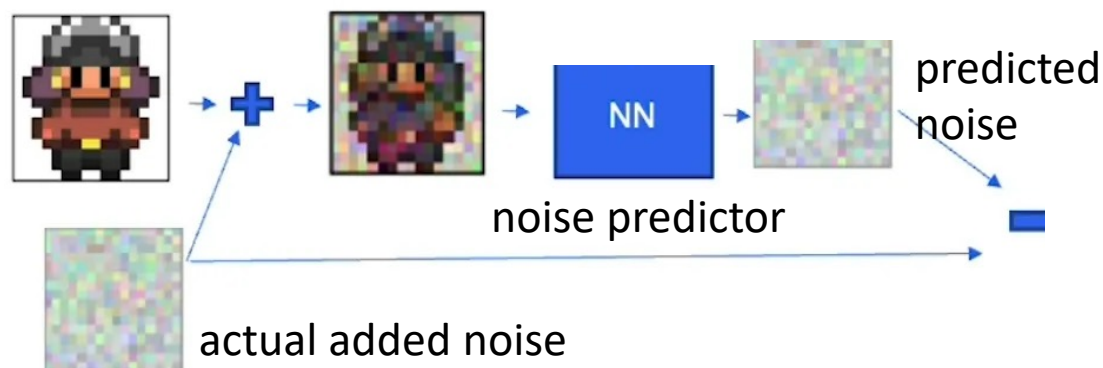
- corrupt a training image by adding a noisy image for T steps (~50-100)
- compare “predicted noise” with “actual added noise” to image
- compute the loss and train NN (**noise predictor**) through backpropagation to estimate the noise added at each step on the corrupted image x_t



Data set:

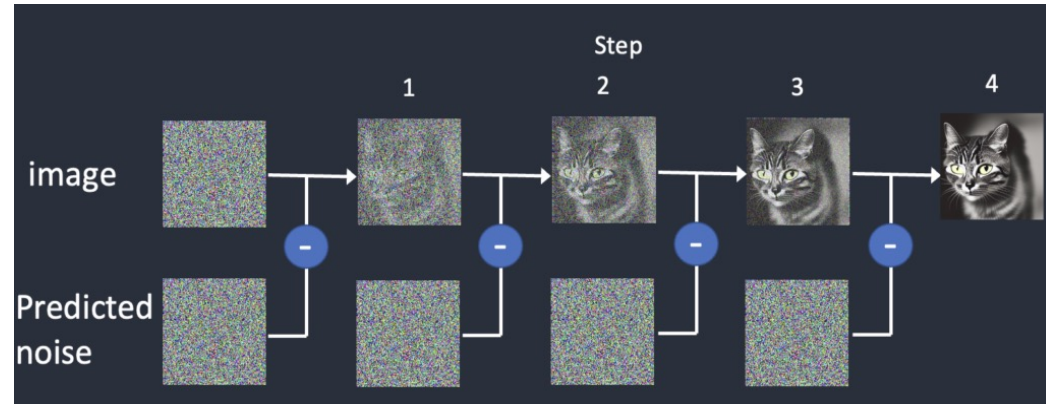
Input - noisy image,
noise amount t

Output - noise sample



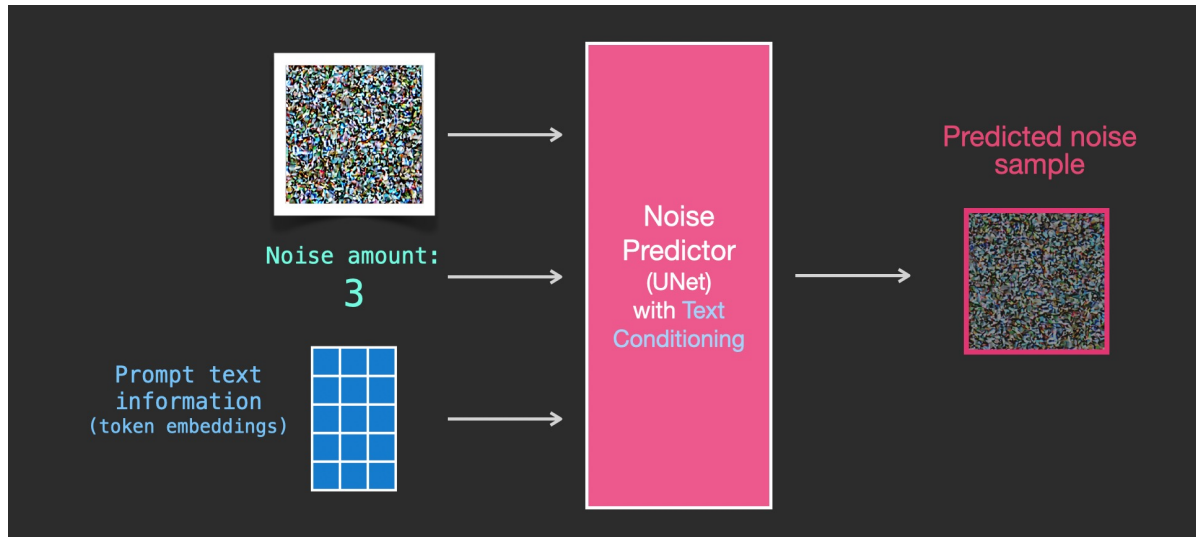
Reverse Diffusion - Generate a image

- **noise predictor** estimates the noise to be subtracted from the original image.
- Repeat this process until you get a cat/dog image
- No control (unconditioned)



To **steer the noise predictor to give what we want** after subtracting.

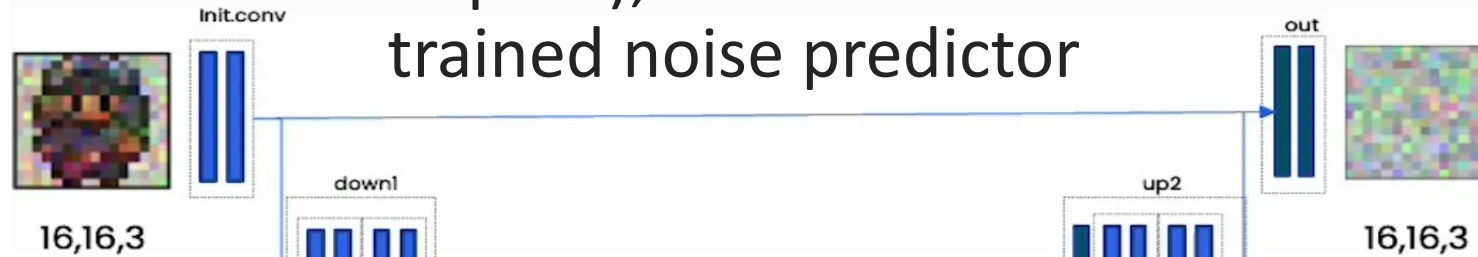
- noise predictor takes text embedding and noisy image to predict noise (Text-to-image)



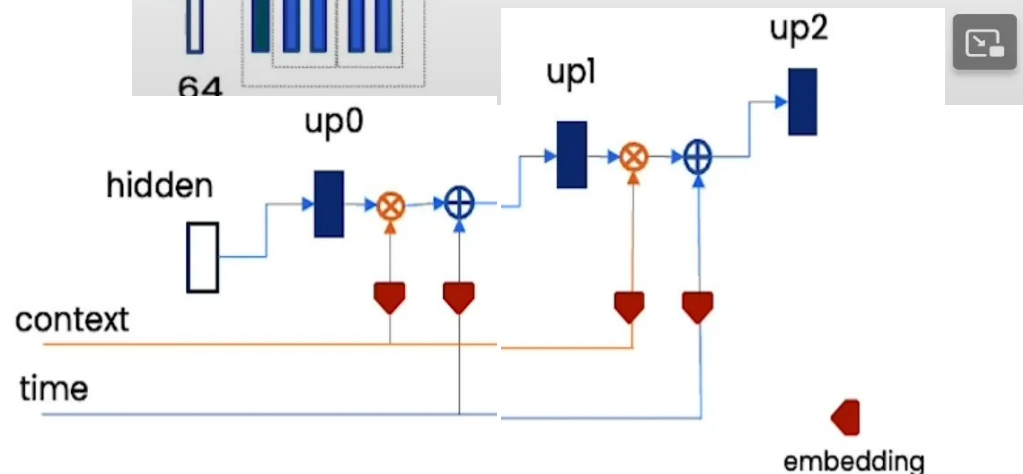
DATASET			
Step	INPUT		OUTPUT / LABEL
	Image	Text	noise sample
3			
14			
7			
42			
2			
21			

Noise Generation - UNet

- Since the shape of input (noisy image) is same as output (predicted noise in each pixel), U-net is used for the trained noise predictor



- Embedding introduces additional vectors to steer the model toward a desired output image.
 - Time embeddings: time steps (noise level)
 - Contextual embeddings: control the generation



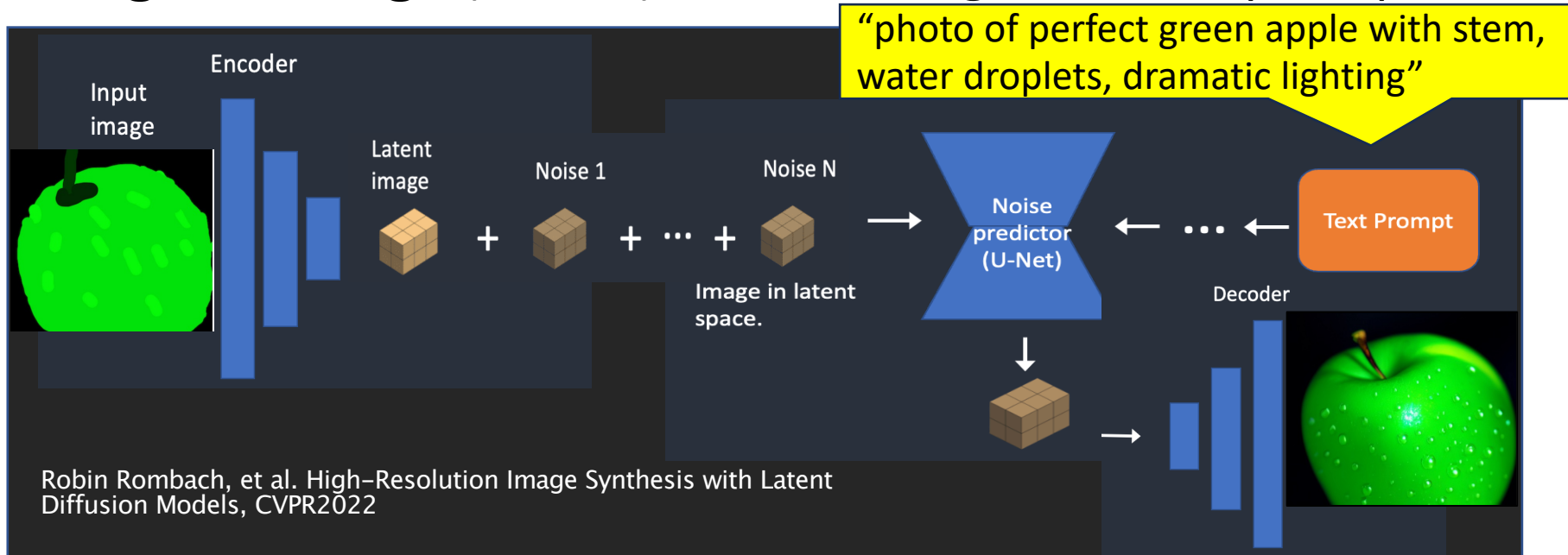
Speed Boost - Latent Diffusion Model

Diffusion process in image space is very slow computationally. To speed up, diffusion process works on latent space of image

The latent space of Stable Diffusion model is $4 \times 64 \times 64$, 48 times smaller than the image pixel space $3 \times 512 \times 512$.

- The input image is **encoded** to latent space. Noise is added/subtracted to/from the latent image
- **Decoder** of VAE converts latent image back to pixel space

Image-to-Image (SDEdit) – latent image and text prompt.



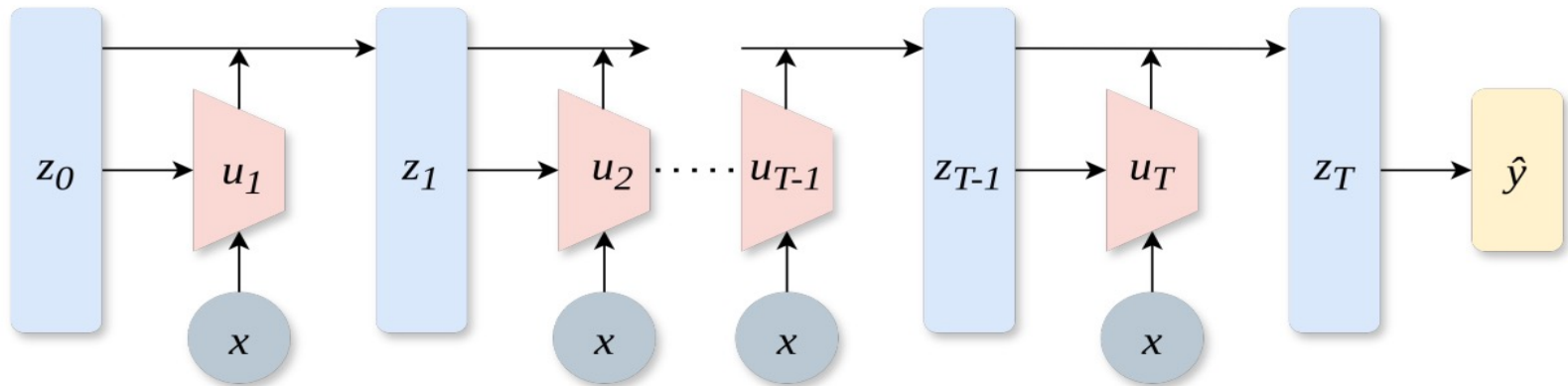
Can we do without BP? (another application)

What's wrong with BP?

[Q. Li, et.al., NoProp: Training Neural Networks without Back-propagation or Forward-propagation, March 2025](#)

- BP is sequential - **cannot run all gradient calculations simultaneously by parallelising them across layers**
- BP is **memory-intensive** - Need to *remember all the outputs produced by each hidden layer during the forward pass*

Proposal: Build a diffusion model that can predict label y given input x , i.e., train a NN to model a stochastic process that transforms random noise into a form that can estimate y (as each layer denoises previous layer's noisy output)



- At each layer, starting from Gaussian noise z_0 , NN u_t sequentially takes input x and noisy z_{t-1} from previous layer (with skip connection), to produce next denoised representation z_t .
- At final step $t=T$, z_T is passed through a classifier to predict final label $\hat{y}$.

Topics

Stable Diffusion

Adversarial Images

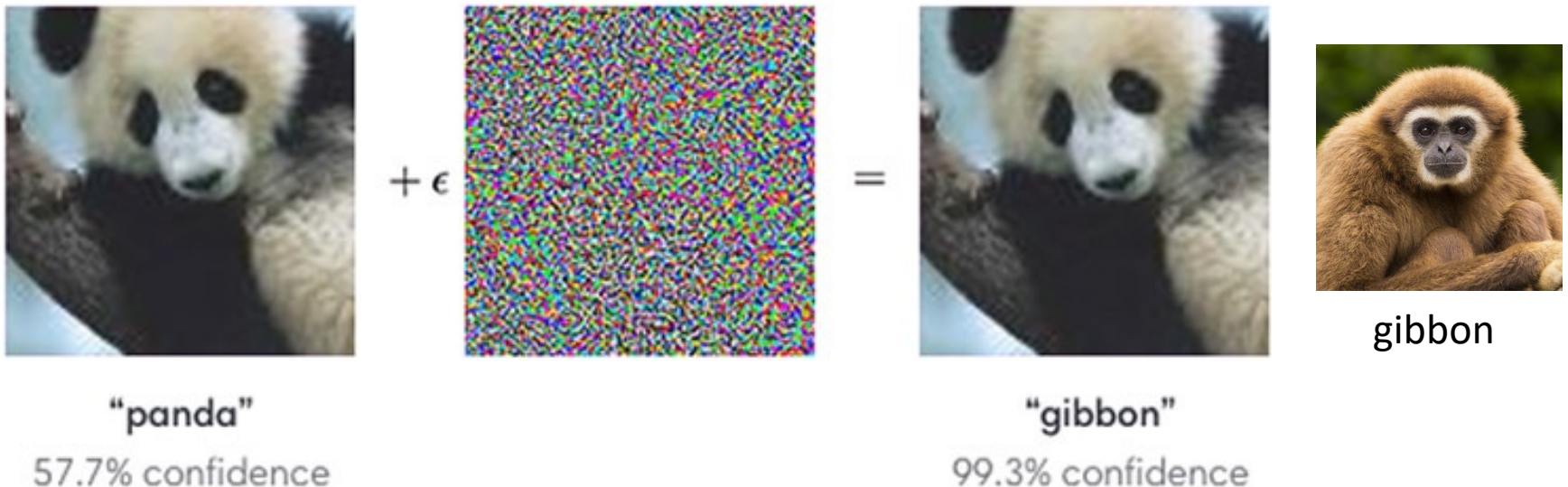
- **Neural Network**
- Other Models

Reflections

- Distributed and Deep representations
- Generalization and Multi-task learning
- Myth of local minima

Adversarial Images

An image specifically designed to fool neural networks into making incorrect classification.

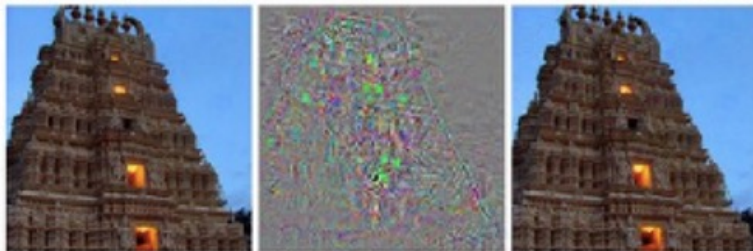
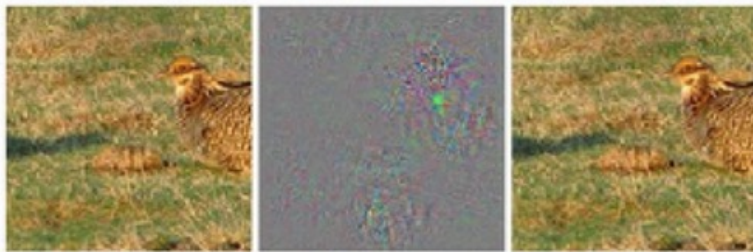
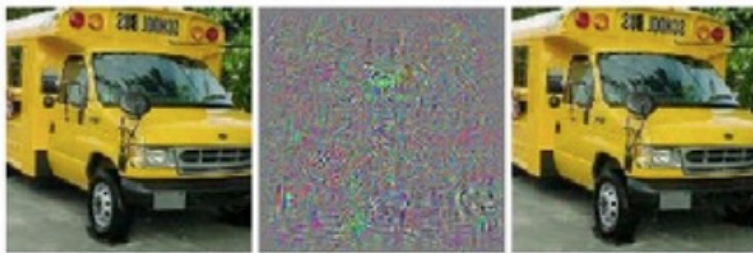


Humans recognize pandas in pictures by abstract features:
little black ears, big white heads, black eyes, fur, ...

CNN works differently.

Adversarial Examples (fooling images)

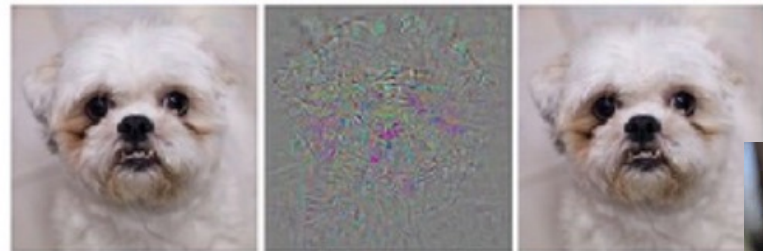
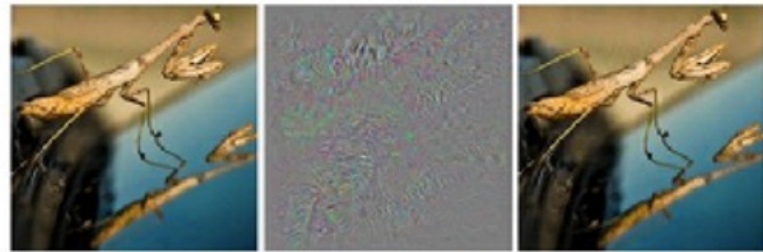
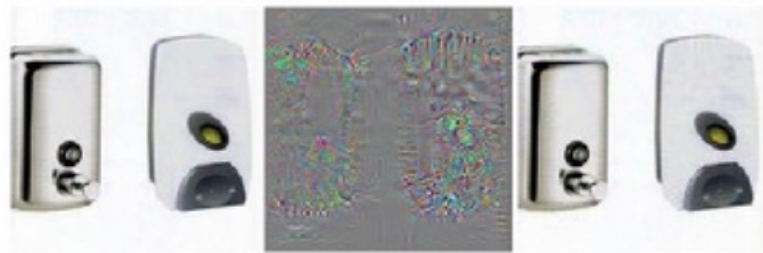
An **adversarial example** is an instance with small, intentional feature perturbations that cause a machine learning model to make a false prediction.



correct

+distort

ostrich



correct

+distort

ostrich



Why interested in Adversarial Examples?

Adversarial examples make machine learning models vulnerable to attacks

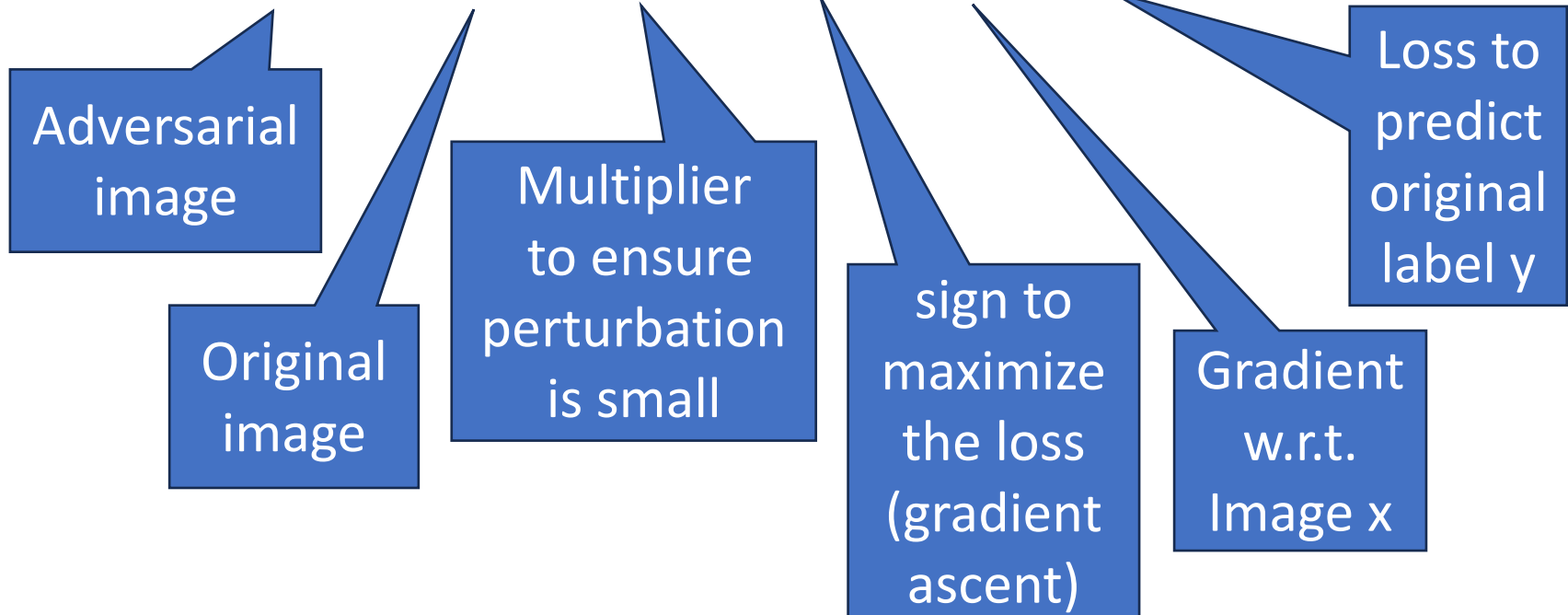
- Self-driving car ignores stop sign (normal to human), which looks like other sign (sign recognition system).
- A spam detector fails to classify an email as spam.
The spam mail has been designed to resemble a normal email, but with the intention of cheating the recipient.
- A machine-learning powered scanner scans suitcases for weapons at the airport.
A knife was developed to avoid detection by making the system think it is an umbrella.

White box attack using FGSM

White box attack is to ensure misclassification.

The **fast gradient sign method** (FGSM) uses the gradients of the loss w.r.t. the input image x to create an adversarial example that **maximizes the loss** J .

$$\text{adv_x} = x + \epsilon * \text{sign}(\nabla_x J(\theta, x, y))$$



Generating Adversarial w Designated Outcome

1. Start from an image x (vector of pixels)
2. Pick a desired outcome class y
3. **Modify the image** by r to maximize class score y or **minimize** the following function with respect to r (with gradient descent) :

$$\text{minimize loss} (f(x+r), y) + c \cdot |r|$$

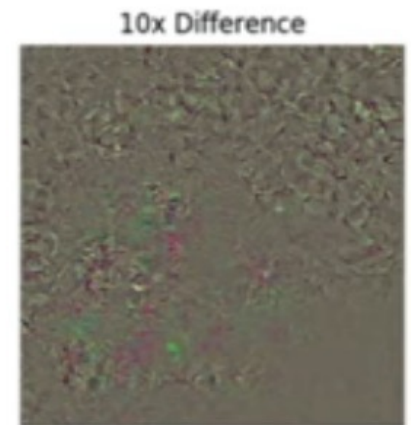
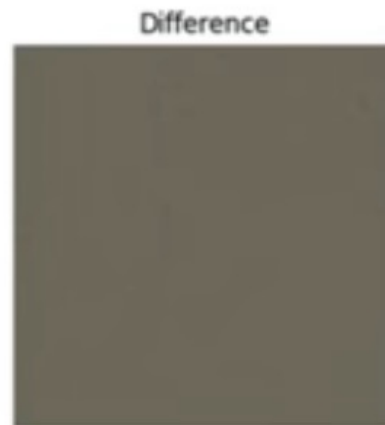
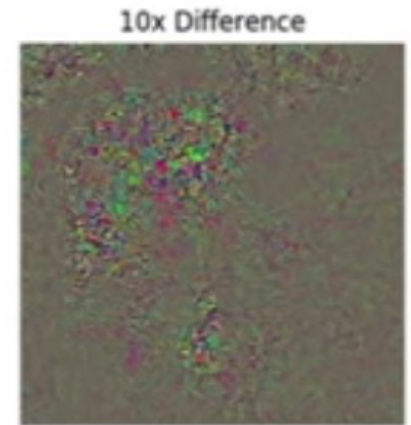
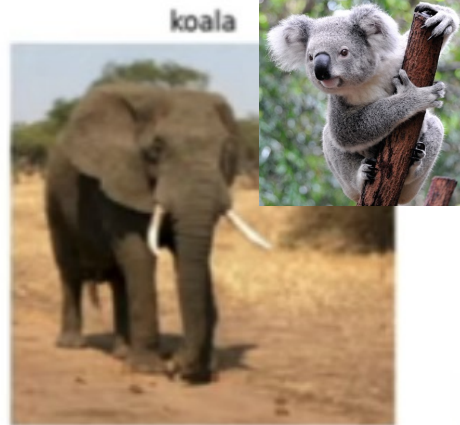
r = changes
made to x

distance between
predicted outcome of
adversarial image ($x+r$)
and the desired class y

$|r|$ = distance
between adversarial
image and the
original image

4. Repeat until network is fooled by $x+r$
(adversary example)

Fooled Images



After very little changes, the elephant and the yacht are classified as koala or iPod, respectively.

Topics

Stable Diffusion

Adversarial Images

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- **Other Models**

Reflections

- Distributed and Deep representations
- Generalization and Multi-task learning
- Myth of local minima

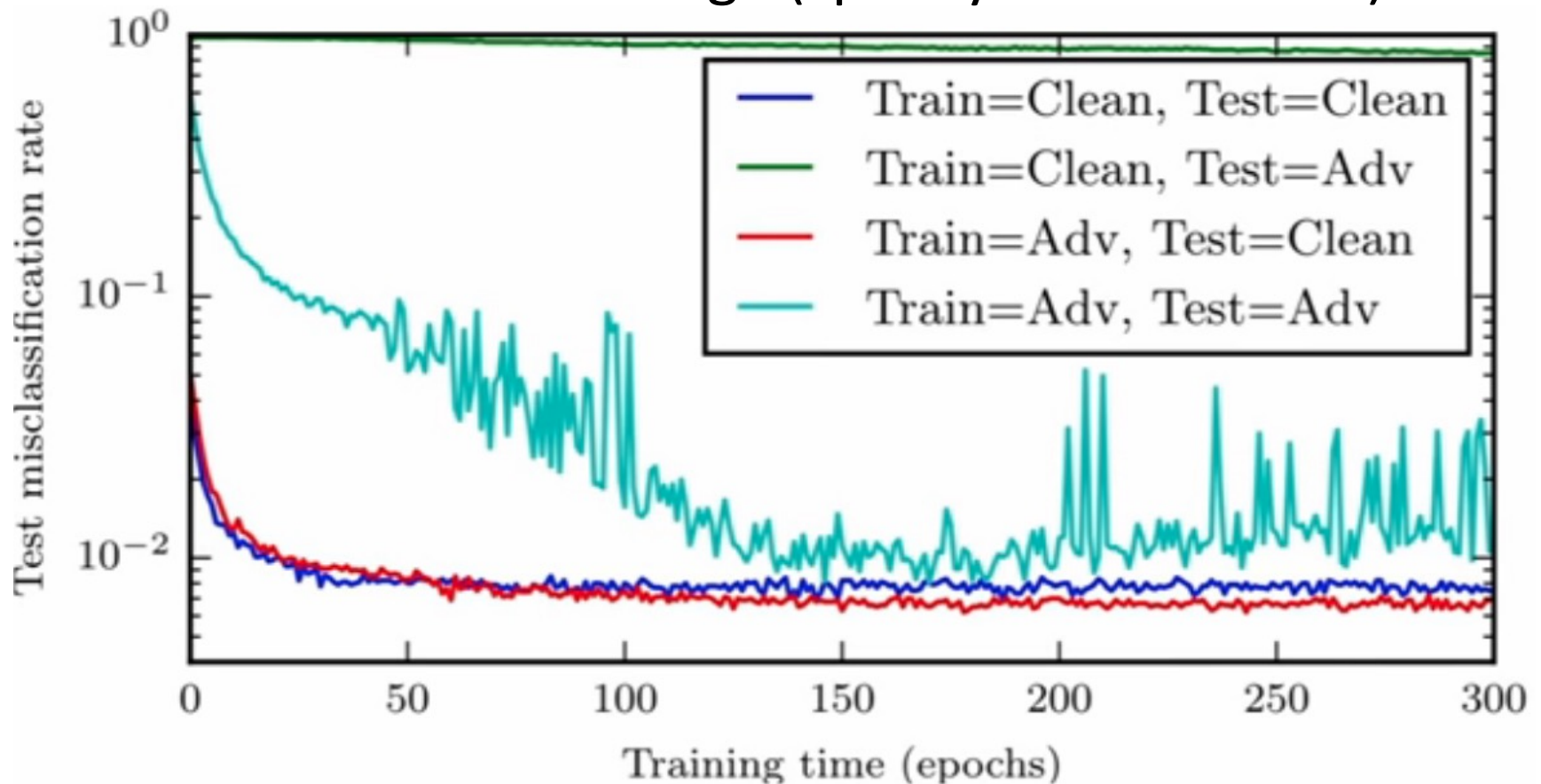
Problem only for Neural Networks?

- Many other machine learning algorithms (e.g., linear models, SVM, k-nn) can be fooled in a similar way.
- Generate an adversarial example from **source** machine learning technique (DT) and use it to fool another classifier (kNN) trained with **target** machine technique.
- The number entry indicates probability being fooled.
The lower the better.
- Cross-technique transferability between deep neural networks, logistic regression, SVM, Decision Trees, k-NN.
- DNN is the best (most difficult to be fooled).

Source Machine Learning Technique \ Target Machine Learning Technique	DNN	LR	SVM	DT	kNN	Ens.
DNN	38.27	23.02	64.32	79.31	8.36	20.72
LR	6.31	91.64	91.43	87.42	11.29	44.14
SVM	2.51	36.56	100.0	80.03	5.19	15.67
DT	0.82	12.22	8.85	89.29	3.31	5.11
kNN	11.75	42.89	82.16	82.95	41.65	31.92

Training with Adversarial Examples

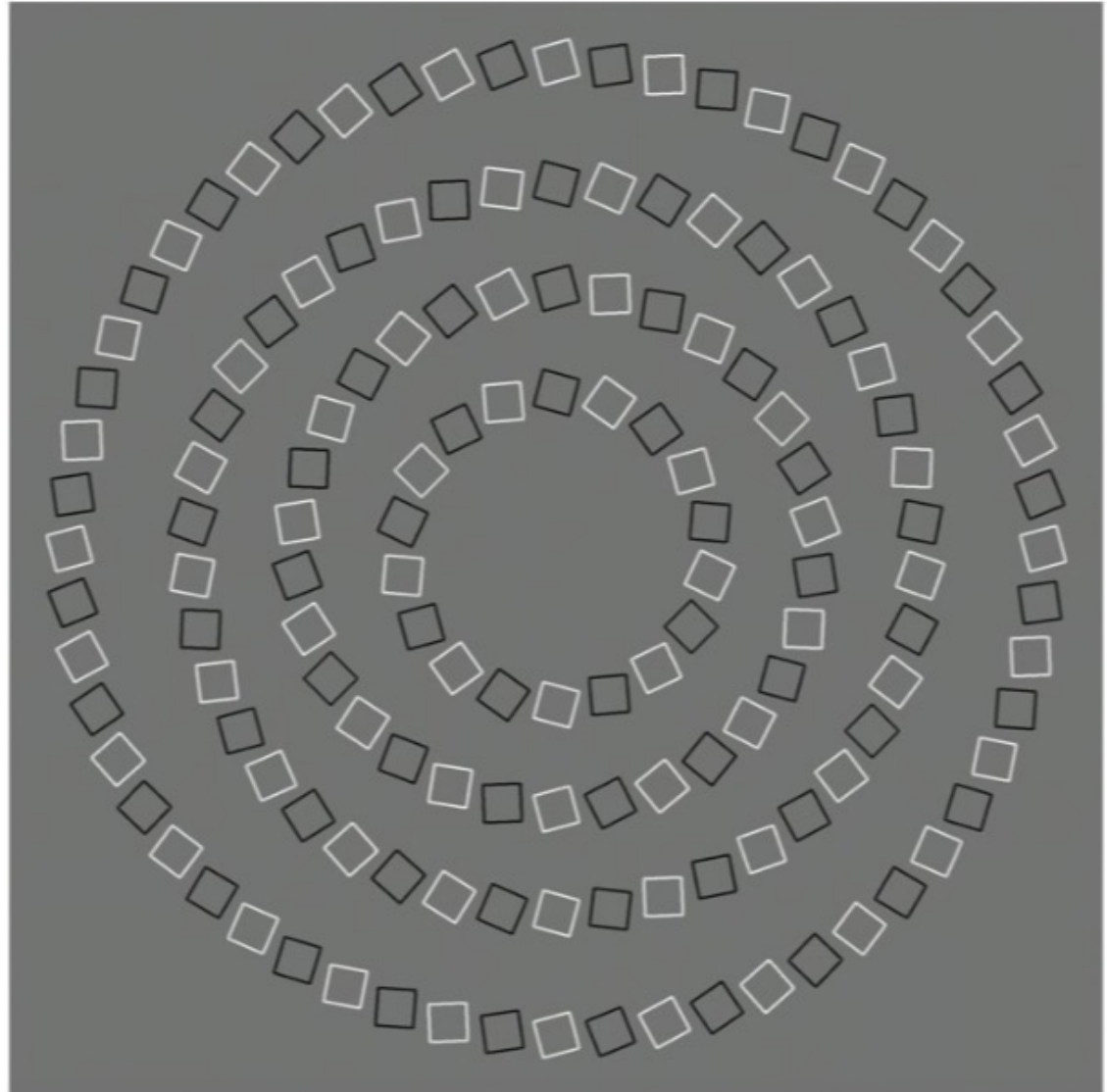
- Training with adversarial examples: for every image add an adversarial image (specify as same class).



- It helps (compare green curve with light blue curve): can resist the same attack you trained on.

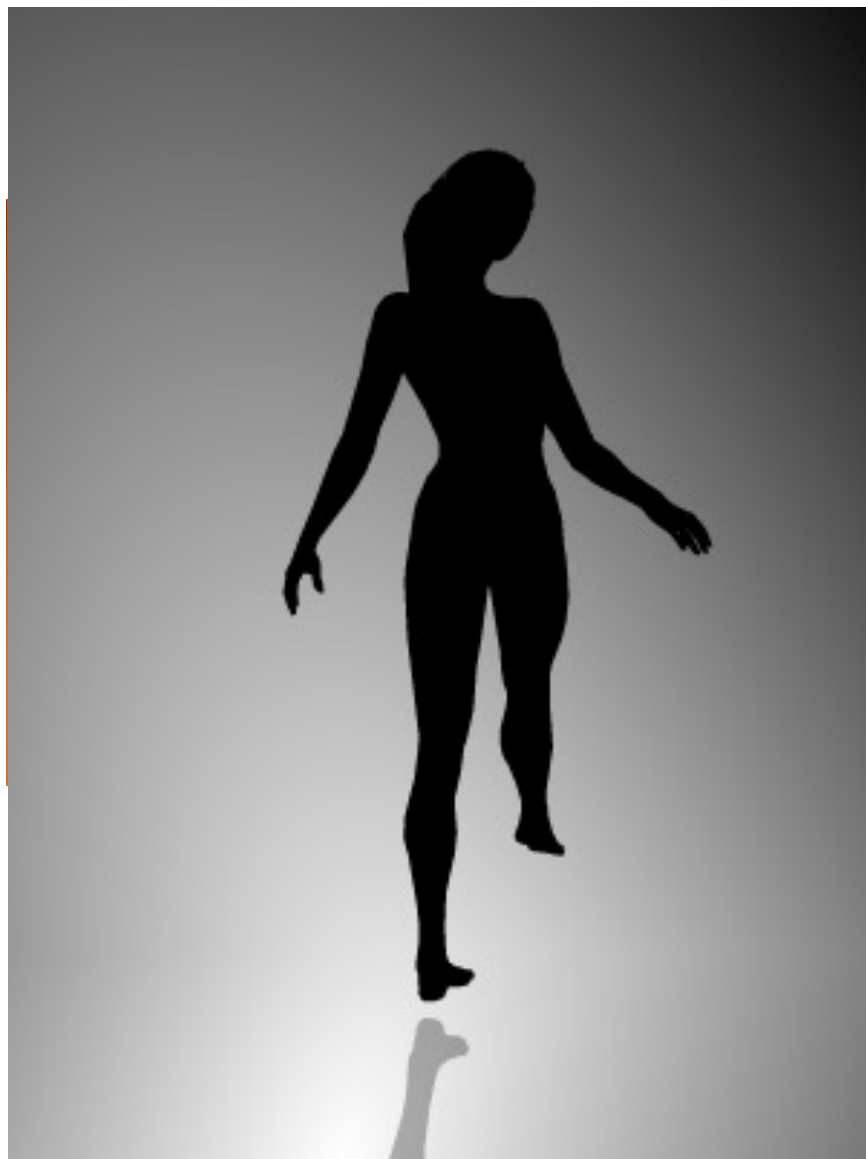
Adversarial Examples for the Human Brain (one of the models)

These are
concentric
circles
and not
intertwined
spirals.



Clockwise or counter-clockwise turn?

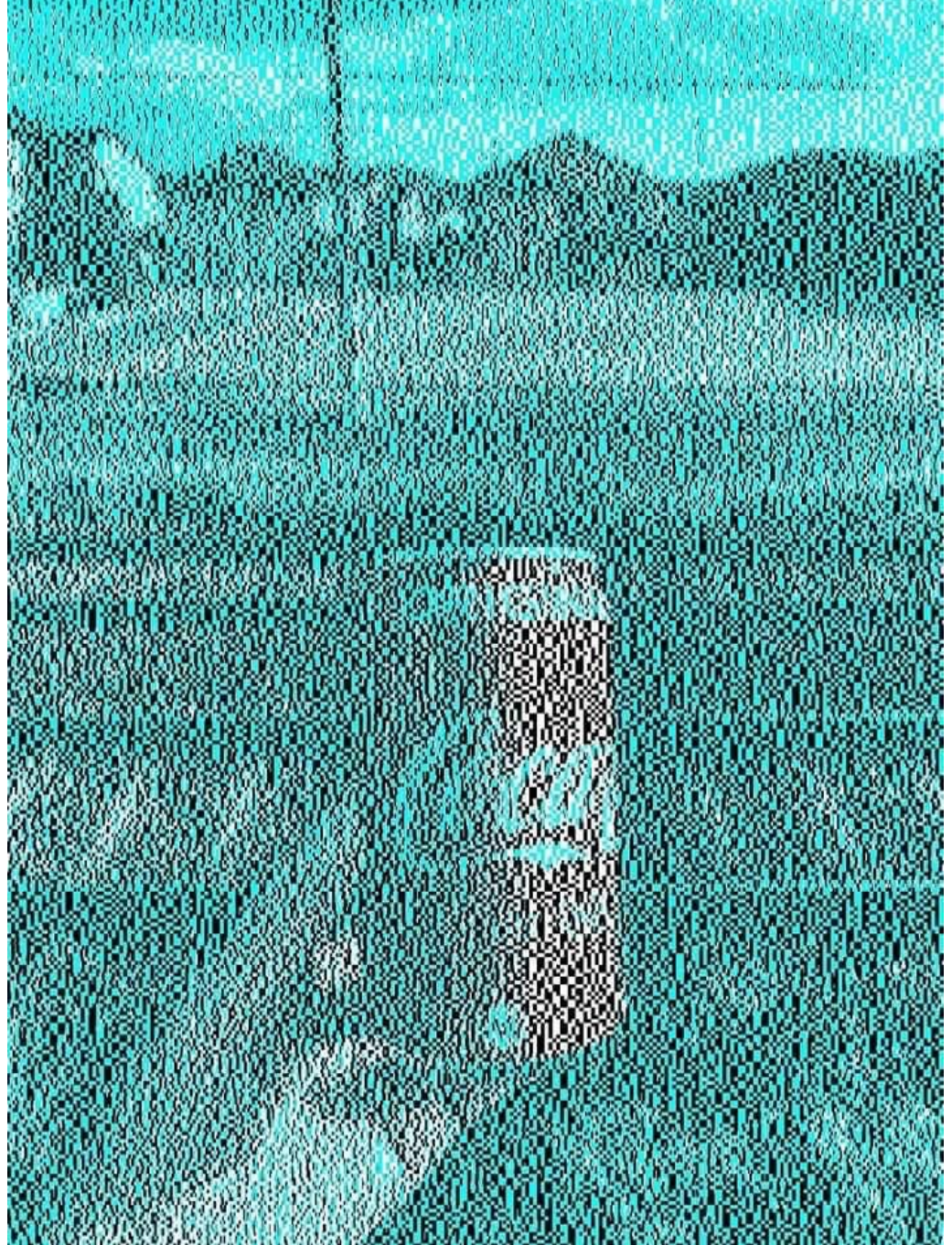
顺时针还是逆时针？



Do you see RED?

The picture is made
entirely of light blue,
black and white.

The brain is filling
in the color.

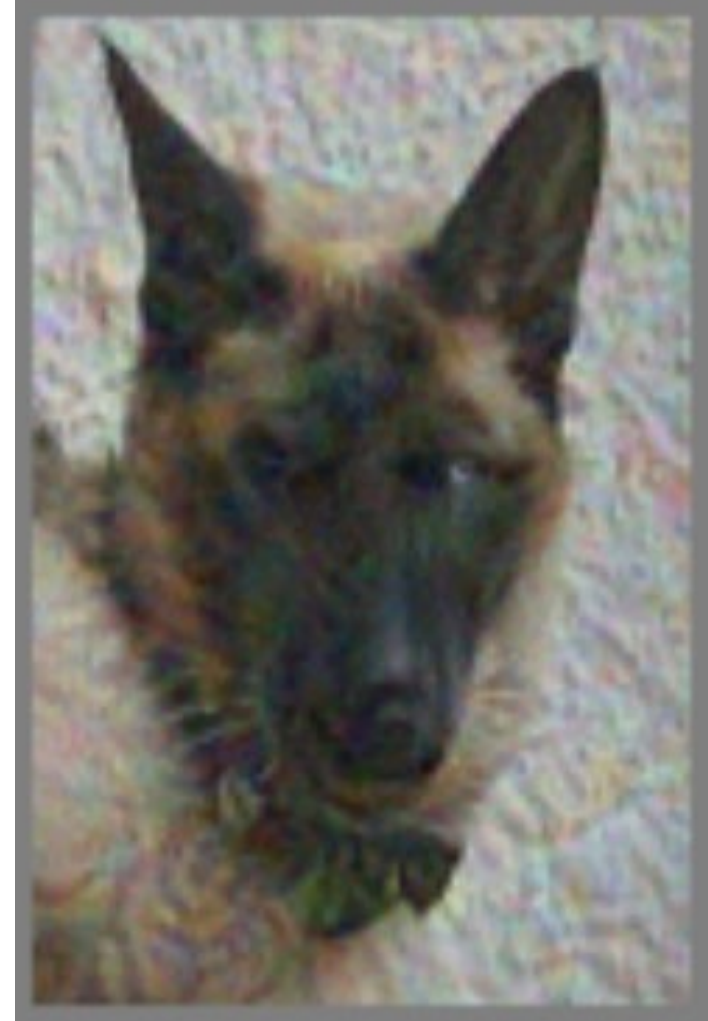
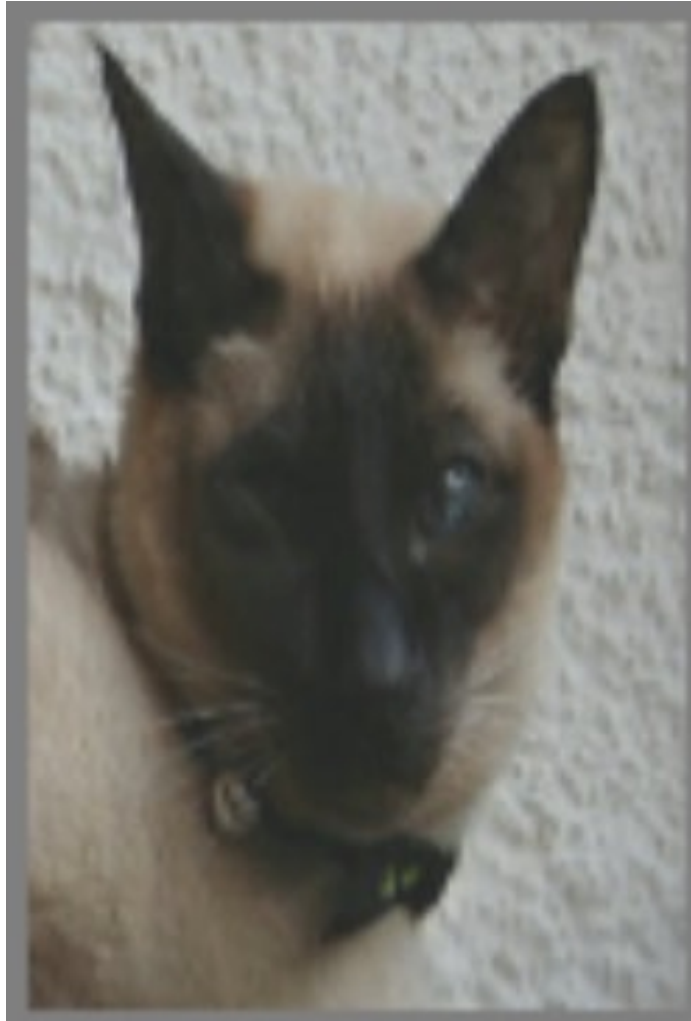


Adversarial Examples for CNN and Human Brain

Left picture is
a CAT

Right picture
(modification
of left picture
slightly)

CNN will
classify
right picture
as a DOG
as well as
many people.



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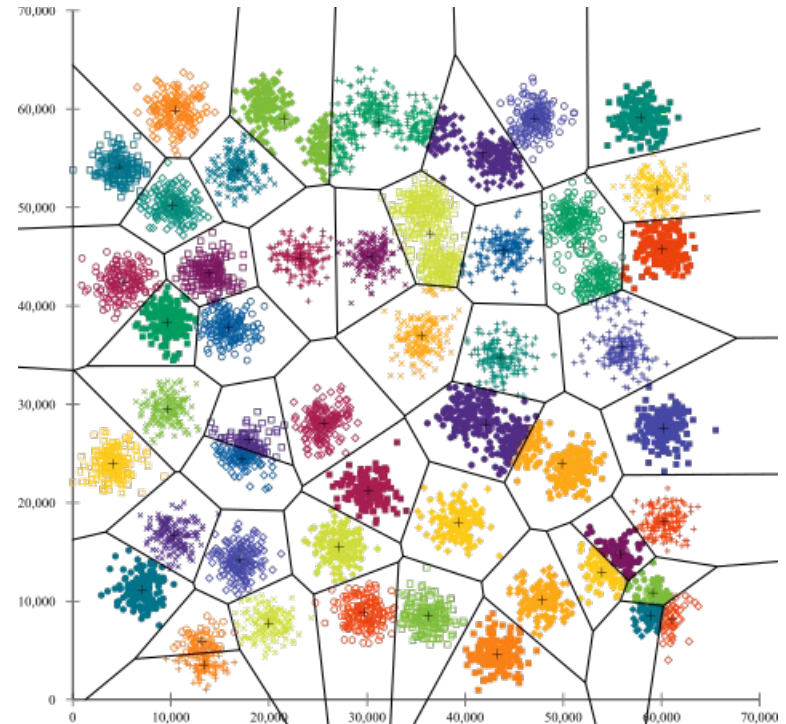
Reflections

- **Distributed and Deep representations**
- Generalization and Multi-task learning
- Myth of local minima

Representations of Data

Non-distributed representations which split space into regions, e.g., clustering, nearest neighbors

- Different parameters for each region
- Complexity = no of regions
- Generalization for unknowns – data vs complexity
- Rich function requires more regions and data
- Number of regions, data and complexity are linearly related
- Symbolic representation e.g., cats and dogs, 2 different concepts, either exist or not with no relationship between them

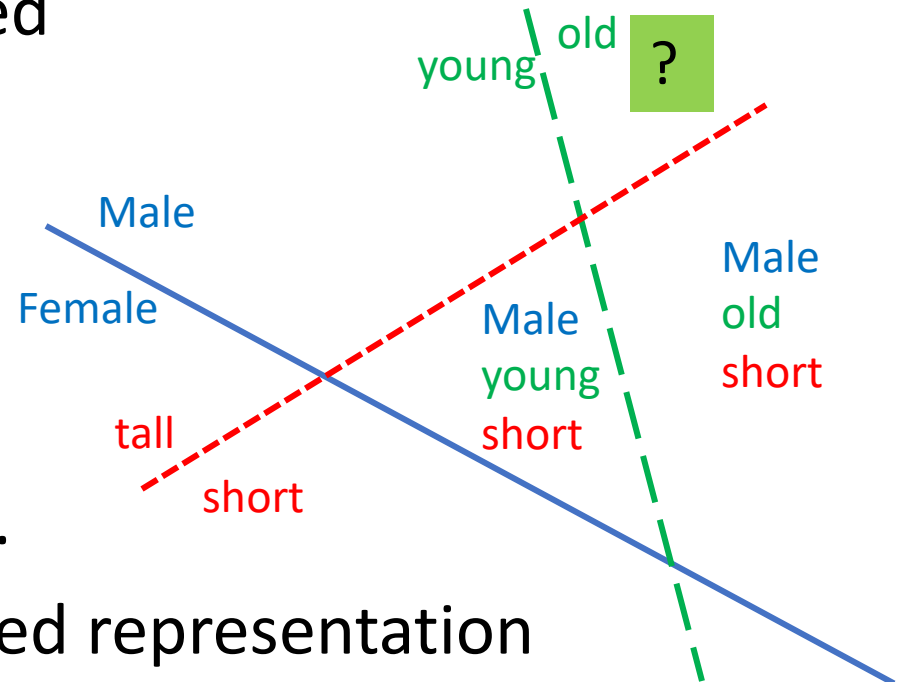


Distributed Representation

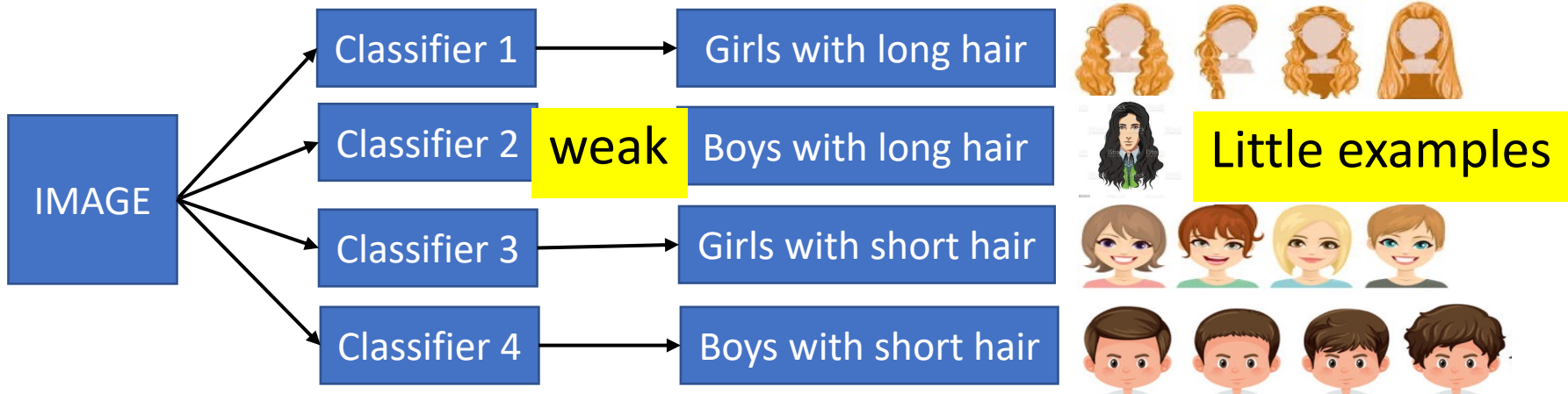
Features are individually meaningful and learned independently despite what the other features are.

Examples: person (sex, height, glasses, race, hair, ...)

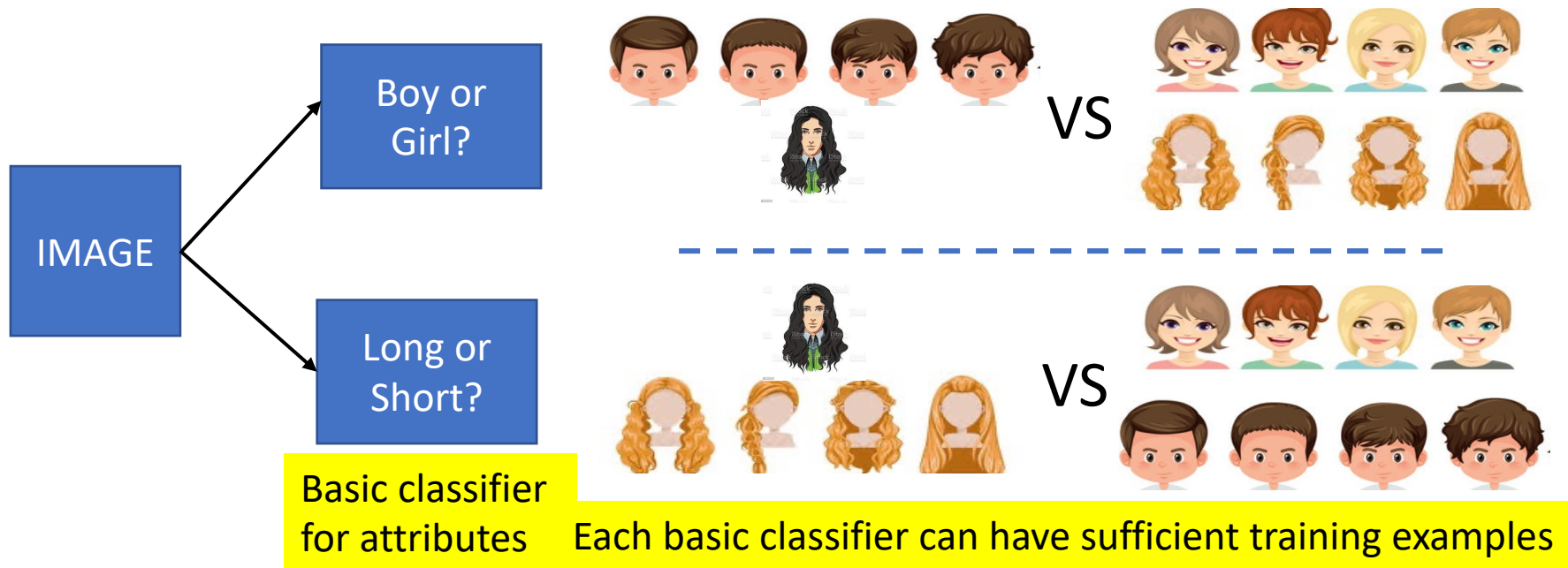
- can represent exponential number of regions with a linear number of parameters
- can learn a very complicated function with a low number of examples
- can deduce something meaningful about what we have not seen before (essence of generalization).
- NLP – symbolic vs distributed representation



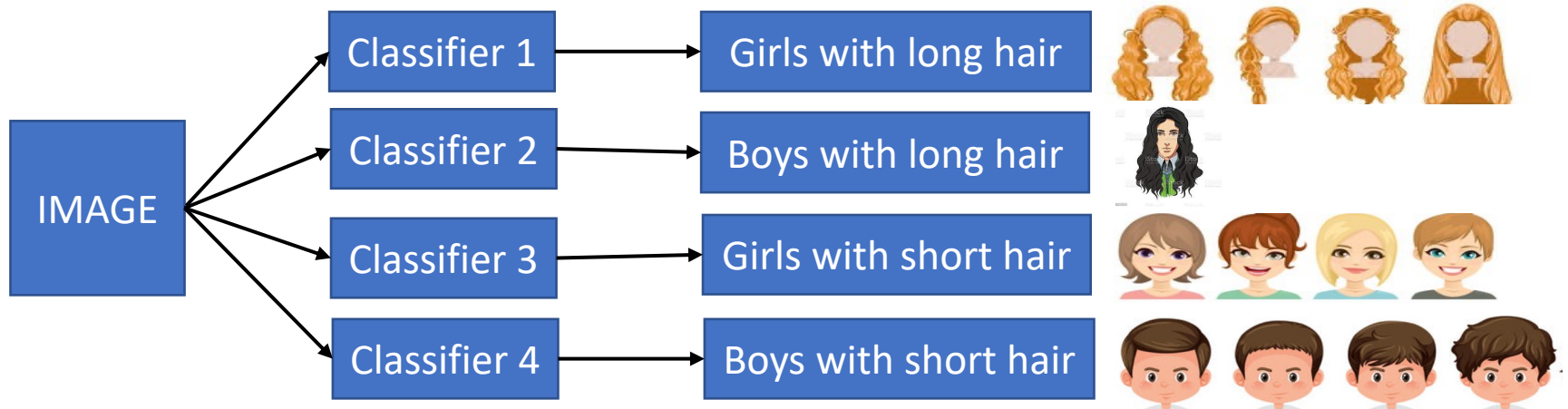
Modularization



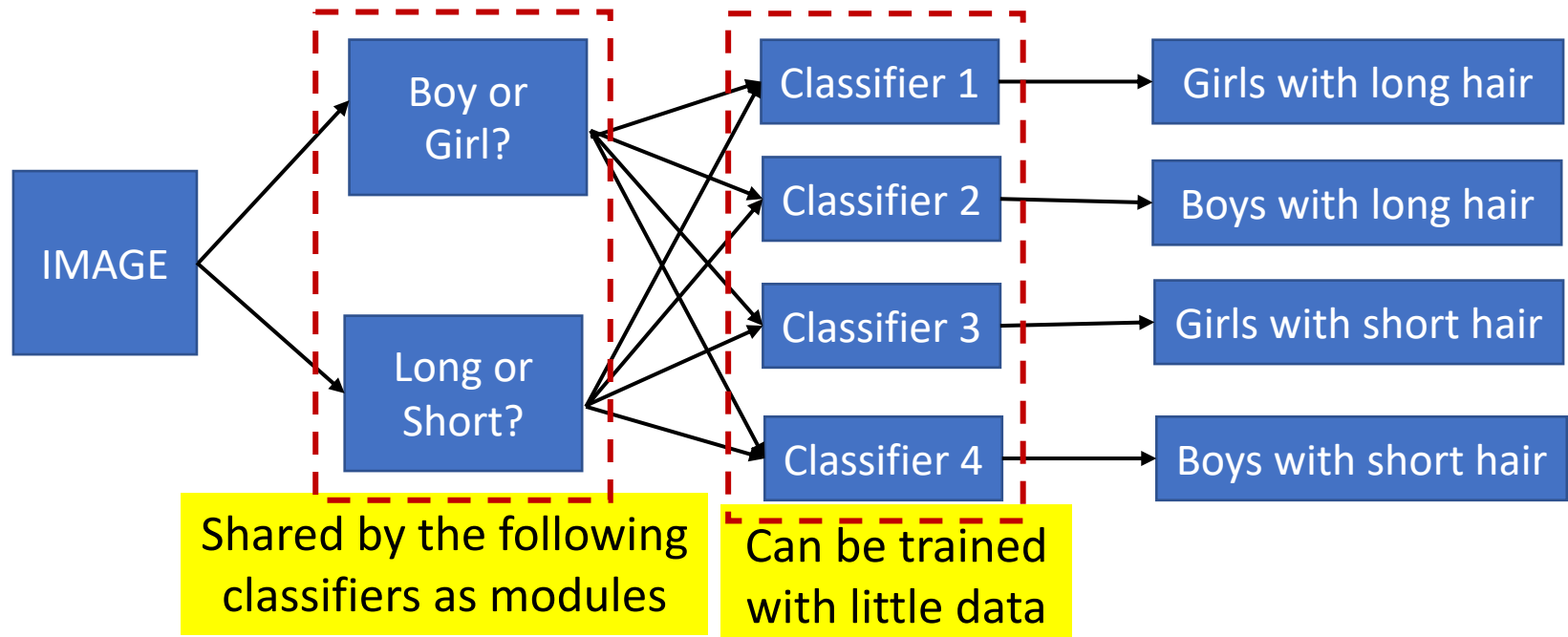
Deep → Modularization



Modularization

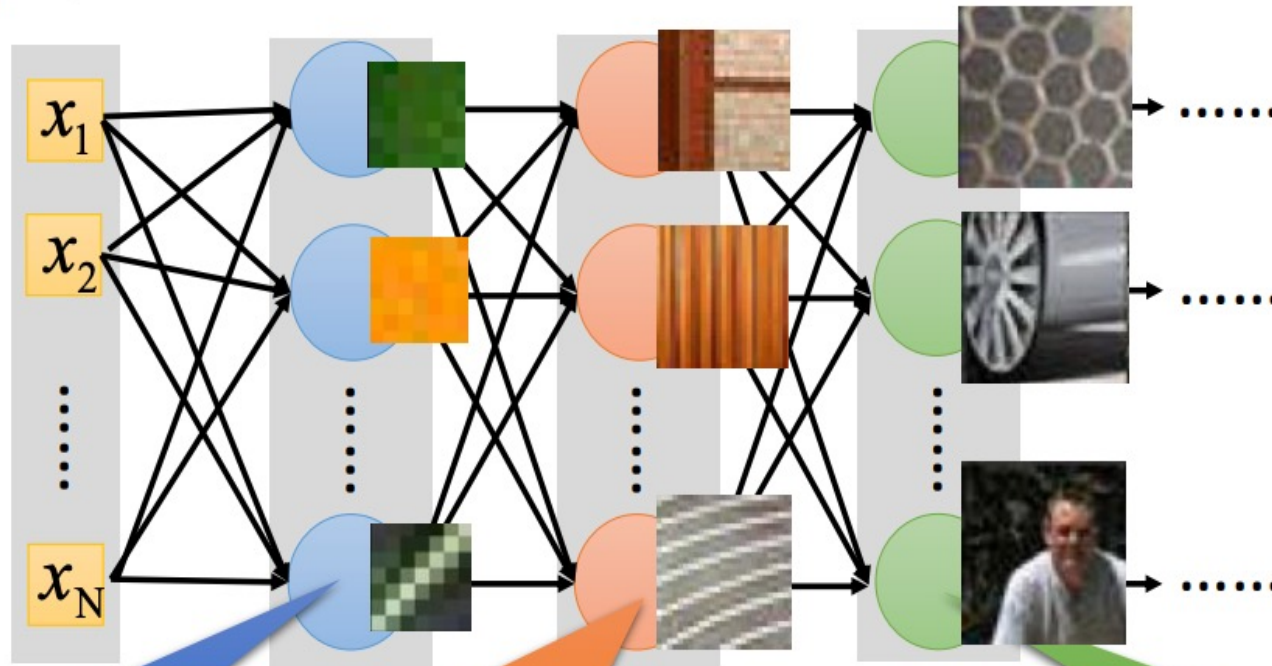


Deep → Modularization



Modularization - Image

- Deep $\rightarrow$ Modularization



The most basic
classifiers

Use 1st layer as module
to build classifiers

Use 2nd layer as
module

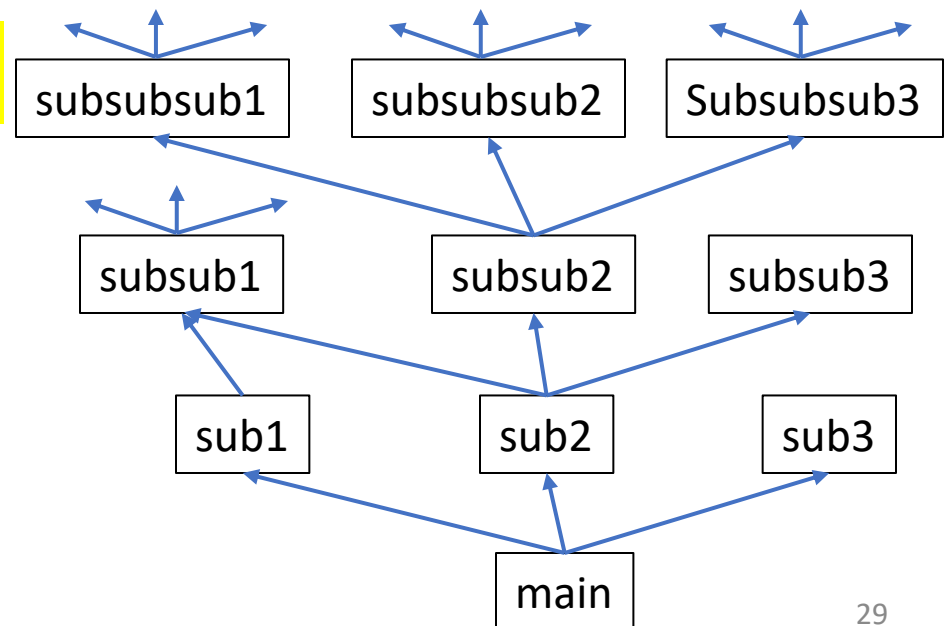
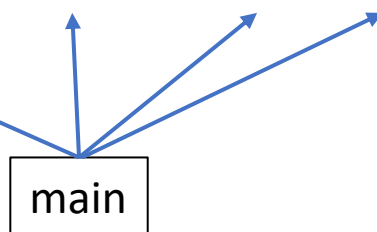
Deep Representation

- Neural network with two layers of hidden units is sufficient to represent any function
- Deep neural network can represent the same function but cheaper, i.e., lesser number of neurons, more effective and simpler (less parameters).
- Deep neural network is not only more powerful, but preferred if the function is composed of many operations (~ shallow/deep computer program)

Fat + Short v.s. Thin + Tall

sub1 code +
subsub1 +
subsub 2 +...

sub2 code +
subsub2 +
subsub 3 +...



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- **Generalization and Multi-task learning**
- Myth of local minima

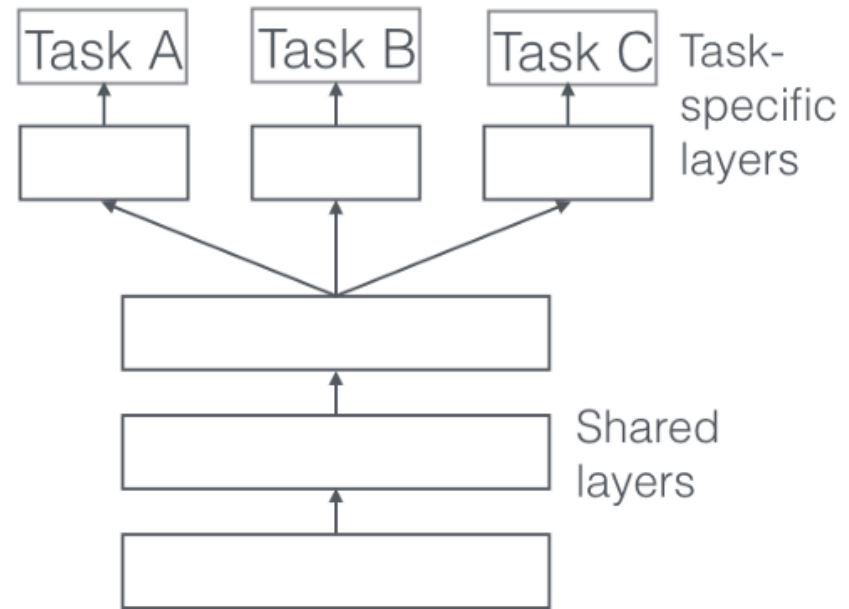
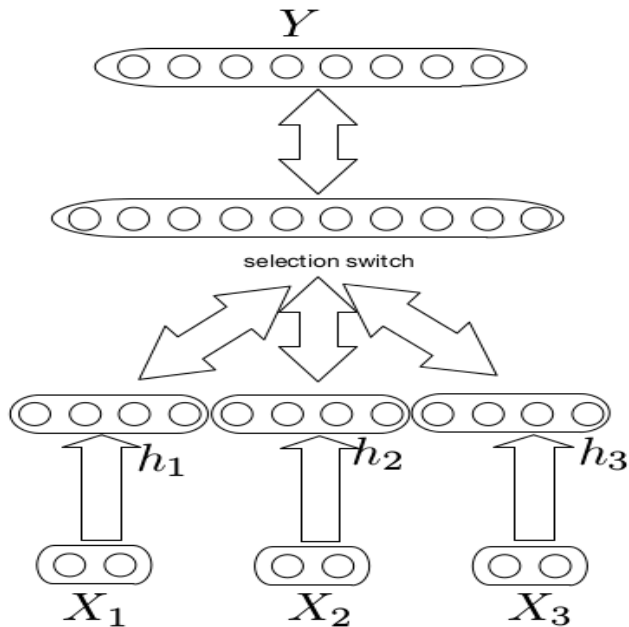
How do children learn?



“Human beings **learn concepts starting from raw data.**
For instance, a child shown a few examples/ instances of
a particular animal (like, say, cats) will soon learn to identify cats.

Multi-task Learning with Different Inputs/Tasks

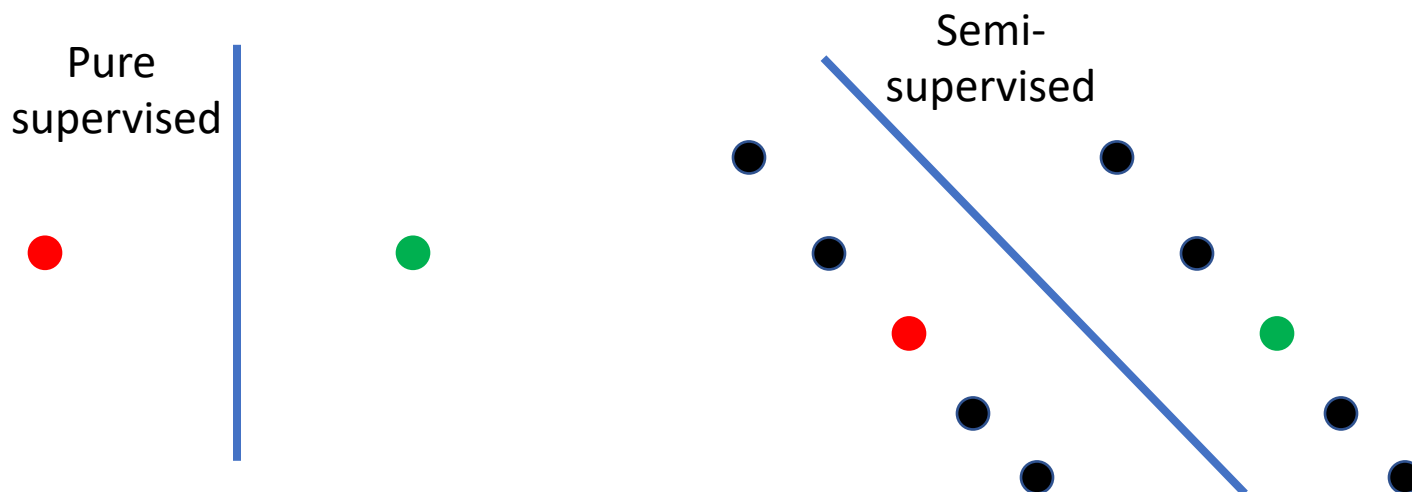
- Generalize from very few examples, sometimes only one (not possible statistically)



- One possibility is to use knowledge from previous learning (prior knowledge) to build representations such that it is possible to generalize from a single example.
- Deep architectures learn good intermediate representations that can be shared across tasks.

Semi-supervised Learning with Deep Distributed Representations

Semi-supervised learning finds a better boundary of classification with unlabeled examples (priors)



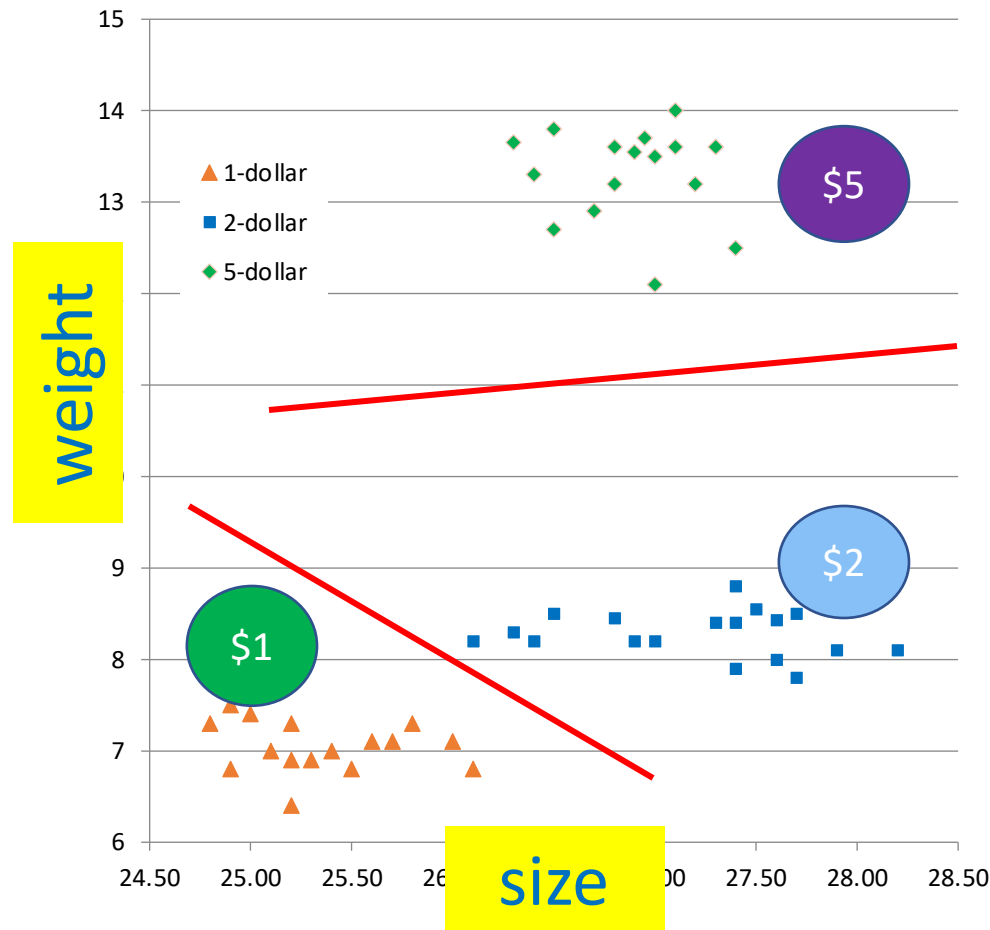
Priors like [semi-supervised learning](#) and [multi-task learning with deep distributed representation](#) can be used in better generalization.

Example on Unsupervised Learning - Coins Classification



Coin Classification

Training Data: **Supervised learning**
($\langle \text{size}, \text{weight} \rangle$, type)

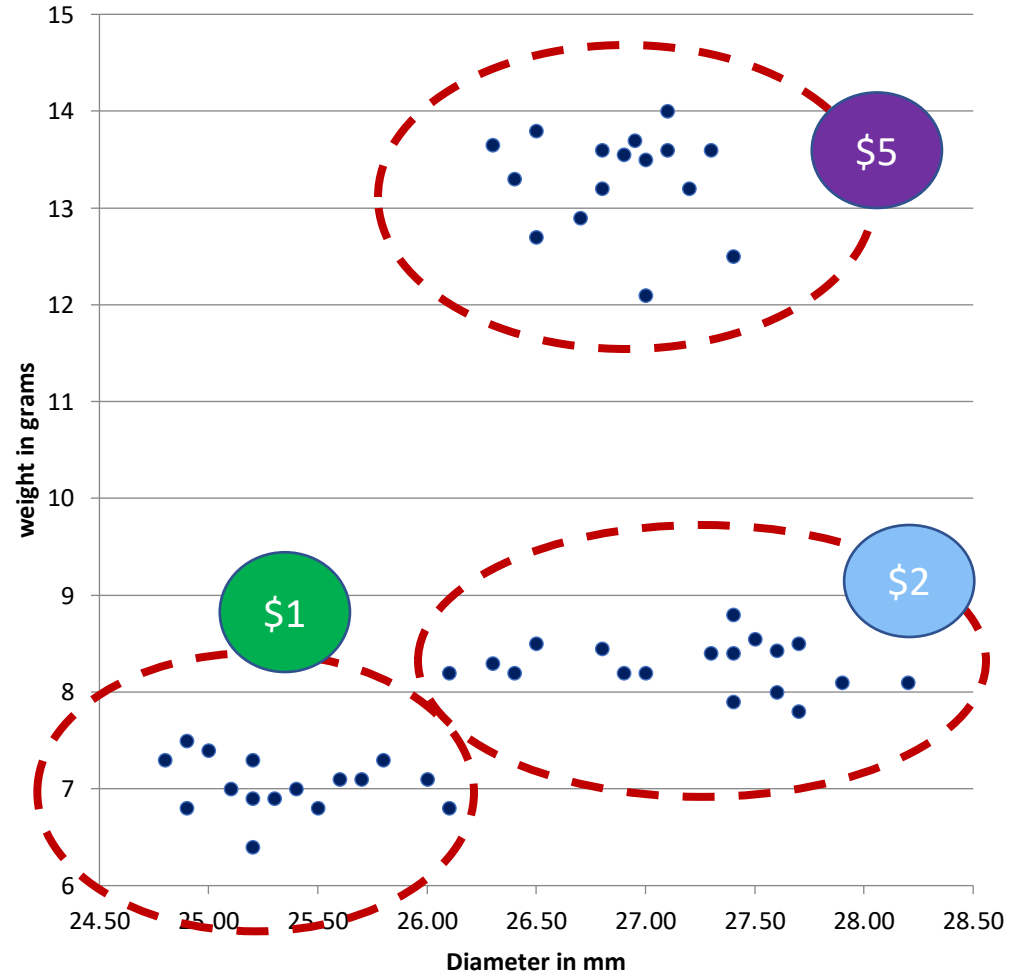


Unsupervised Learning

Grouping exists

Learn from raw data about patterns to classify \$1 and \$5

Can learn how to classify \$2 with one example in the group



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- **Myth of local minima**

Myth of Local Minima

- Difficulties in getting optimal solution
 - Functions usually non-convex
 - Exponential number of local minima (multi-dimensions)
- Local minimum occurs when all directions go up in all dimensions
- Probability of local minimum $\sim 2^{-n}$ exponentially unlikely (with n = no. of dimensions)
 - many saddle points.
- Local minimum usually occurs when near the bottom of landscape, i.e., global minimum.

